Some Integral Identities Formed from Differentiating the Laplace Transform

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To give a general flavor of what I mean by "differentiating the Laplace Transform", let us use the identity $\mathcal{L}[t^k f(t)] = (-1)^k \frac{d^k}{ds^k} F(s)$, found in any table relating a given function $f(t)$ to its Laplace Transform $F(s) = \int_0^\infty e^{-st} f(t) dt$, to derive the Taylor Series:

$$F(a) = \int_0^\infty e^{-st} e^{(s-a)t} f(t) dt = \int_0^\infty e^{-st} \sum_{k=0}^\infty \frac{(a-s)^k (-t)^k}{k!} f(t) dt = \sum_{k=0}^\infty \frac{(a-s)^k}{k!} \int_0^\infty e^{-st} (-t)^k f(t) dt = \sum_{k=0}^\infty \frac{(a-s)^k}{k!} F^{(k)}(s)$$

Which gives us an expansion of $F(a)$ in terms of its derivatives at the base point s. It is this sort of well-known fluidity with which derivatives of the Laplace Transform can be translated into "simpler" operations of the input $f(t)$ which form the theme for this miscellaneous note.

Constructing Log-Product Integrals

There is a most beautiful method of transforming a product of rational functions $\prod_n \frac{p(n)}{q(n)}$ into an integral identity or formula by finding a function $f(t)$ whose Laplace Transform is the quantity $F'(s) = \frac{p'(s)}{p(s)} - \frac{q'(s)}{q(s)}$ and then "integrating" this quantity using the classic identity $\mathcal{L}[t^{-1} f(t)] = \int_s^\infty F'(u) du = -F(s)$ to arrive at $\ln p(s) - \ln q(s)$ which can then be summed back into a "log-product" $\ln\left(\prod_n \frac{p(n)}{q(n)}\right)$. That's the idea at least, but it's much easier to see with some concrete examples!

The famous *Euler Product* for sine takes the form $\prod_{n\geq 1}\left(1 - \frac{\alpha^2}{n^2}\right) = \frac{\sin(\pi\alpha)}{\pi\alpha}$ and so if we'd like to construct an integral "around" this identity, we'd like to find a Laplace transform $F(s) = \ln\left(1 - \frac{\alpha^2}{s^2}\right)$. If successful, we could then use $\mathcal{L}[f(t)] = F(s)$ to arrive at;

$$\ln\left(\frac{\sin(\pi\alpha)}{\pi\alpha}\right) = \sum_{n=1}^\infty \ln\left(1 - \frac{\alpha^2}{n^2}\right) = \sum_{n=1}^\infty \int_0^\infty e^{-nt} f(t)\, dt = \int_0^\infty \frac{e^{-t}}{1-e^{-t}} f(t)\, dt$$

But finding $f(t)$ from $F(s)$ is far clearer after taking the derivative $F'(s) = \frac{2s}{s^2-\alpha^2} - \frac{2}{s}$ which, thanks to being a linear combination of rational functions, can be easily looked up in a Laplace Transform table. In short, $f(t) = 2\cosh(\alpha t) - 2$ does the trick. So, upon applying the $\mathcal{L}[t^{-1} f(t)]$ identity from above, we conclude $\ln\left(1 - \frac{s^2}{\alpha^2}\right) = 2\int_0^\infty e^{-st}\left[1 - \cosh(\alpha t)\right]\frac{dt}{t}$ which yields;

$$\ln\left(\frac{\sin(\pi\alpha)}{\pi\alpha}\right) = \sum_{n=1}^\infty 2\int_0^\infty e^{-nt}\cdot\left[1-\cosh(\alpha t)\right]\frac{dt}{t} = 2\int_0^\infty \left[1-\cosh(\alpha t)\right]\frac{1}{t}\frac{1}{e^t-1} dt$$

Interestingly, in the American Mathematical Monthly problem *#12338* I. Mezo invites the reader to prove $\int_0^\infty \frac{1-\cos(t)}{t(e^t-1)} dt = \frac{1}{2}\ln\left(\frac{\sinh\pi}{\pi}\right)$, which is precisely the above identity at $\alpha = i$: recall that $\cosh(it) = \cos(t)$ and $\sin(i\pi) = i\sinh(\pi)$. Perhaps Mezo's construction of the problem involved a procedure identical to ours.

> *Exercise:* The above log-sine identity converges only for certain values of α: what are they? For what values of x does $\frac{1}{1-x} = -\sum_{n=1}^\infty x^{-n}$ or $\frac{1}{1-x} = \sum_{n=0}^\infty x^n$ converge? Why is this interesting?

Notice the *characteristic* suffix $\frac{1}{t}\frac{1}{e^t-1}$: the $\frac{1}{e^t-1}$ term is the result of summing the geometric series in e^{-nt} while the $\frac{1}{t}$ term is the result of integrating the Laplace transform $F'(s)$. We'll see this pattern again in our next example:

$$\ln\binom{n}{k} = \int_0^\infty \left(1-e^{-kt}\right)\left(1-e^{-(n-k)t}\right)\frac{1}{t}\cdot\frac{1}{e^t-1}dt$$

Which is beautifully invariant under swapping $k \mapsto n-k$, just like the binomial on the LHS.

Now, the most straightforward way of turning the binomial coefficient into a log-rational function would probably be to select $F'(s) = \frac{1}{s-\alpha} - \frac{1}{s+1}$ and this is because;

$$\sum_{j=0}^{k-1} F(j) = \ln(\alpha-j) - \ln(j+1) = \ln\frac{(\alpha-0)\times(\alpha-1)\times\ldots(\alpha-k+1)}{1\times 2\times\ldots k} = \ln\binom{\alpha}{k}$$

Gives us a nice finite sum over the k elements in the product $\binom{\alpha}{j} = \prod_{j=0}^{k-1}\frac{\alpha-j}{j+1}$. This choice of $F'(s)$ certainly is not an invalid approach, since by recognizing $f(t) = e^{\alpha t} - e^{-t} = \mathcal{L}^{-1}[F'(s)]$ we can quickly deduce;

$$F(s) = \int_0^\infty e^{-st}\frac{f(t)}{t}dt \implies \ln\binom{\alpha}{k} = \int_0^\infty \sum_{j=0}^{k-1} e^{-jt}\frac{e^{-\alpha t}-e^{-t}}{t}dt = \int_0^\infty \frac{e^{-kt}-1}{e^{-t}-1}\frac{e^{-\alpha t}-e^{-t}}{t}dt$$

However the issue is that this integral only converges for $\alpha < 0$. To overcome this, an infinite series in j is employed together with a telescoping structure for $F'(s) = \left[\frac{1}{s+\alpha-k} - \frac{1}{s+\alpha}\right] - \left[\frac{1}{s} - \frac{1}{s+k}\right]$. When we work out the cancellation of terms we find that only the desired k stick around;

$$\sum_{j=1}^{\infty} F(j) = [\ln(j+\alpha-k)-\ln(j+\alpha)]-[\ln(j)-\ln(j+k)] = [\ln(\alpha+k-1)+\ln(\alpha+k-2)+\ldots\ln(\alpha)]-[\ln(1)+\ln(2)+\ldots\ln(k)] = \ln\binom{\alpha}{k}$$

And leave it to the interested reader to check that the tail of the series actually vanishes to 0. Once again, the simple rational structure for $F'(s)$ makes finding the corresponding $f(t) = e^{-(\alpha-k)t} - e^{-\alpha t} - 1 + e^{-kt} = -(1-e^{-kt})(1-e^{-(\alpha-k)t})$ very simple, and so we quickly deduce $\ln\binom{\alpha}{k} = -\sum_{j=1}^{\infty}\int_0^\infty e^{-jt}\frac{f(t)}{t}dt$ which gives the claimed identity.

A tightly related construction leads us to a similar-looking identity;

$$\ln(n!) = \sum_{k=1}^{n}\ln(k) = \sum_{k=1}^{n}\int_0^\infty \frac{e^{-t}-e^{-kt}}{t}dt = \int_0^\infty \left(n-1-ne^{-t}+e^{-nt}\right)\frac{1}{t}\cdot\frac{1}{e^t-1}dt$$

Which if we permit $n=\alpha$ to vary as a continuous parameter like we did with the binomial coefficient, one recovers a famous formula of Binet for the *log-Gamma* function: $\ln(\alpha!) = \ln(\alpha\Gamma(\alpha))$.

What's nice about this is that we can now take the above integral and *undo* the Laplace construction to turn $\alpha\Gamma(\alpha)$ back into a product: if $f(t) = \alpha - 1 - \alpha e^{-t} + e^{-\alpha t} = (1-e^{-t\alpha}) - \alpha\,(1-e^{-t})$, then it's Laplace transform ought to be $F'(s) = \left[\frac{1}{s} - \frac{1}{s+\alpha}\right] - \alpha\left[\frac{1}{s} - \frac{1}{s+1}\right]$ and hence;

$$\ln\left(\alpha\Gamma\left(\alpha\right)\right) = -\sum_{n\geq 1}\underbrace{\int_0^\infty f(t)\frac{e^{-nt}}{t}dt}_{-F(n)} = \sum_{n\geq 1}\int_n^\infty \underbrace{\left[\frac{1}{s}-\frac{1}{s+\alpha}\right]-\alpha\left[\frac{1}{s}-\frac{1}{s+1}\right]}_{F'(s)}ds$$

$$= \sum_{n\geq 1}\ln\left(\frac{n+\alpha}{n}\right) - \alpha\ln\left(\frac{n+1}{n}\right) = \ln\prod_{n\geq 1}\frac{1+\frac{\alpha}{n}}{\left(1+\frac{1}{n}\right)^{\alpha}}$$

Which, coming full circle, is known as the *Euler Product* for the Γ function.

Exercise: The *Frullani Integral* is the identity;

$$\int_0^\infty \frac{f(bt)-f(at)}{t}dt = \ln(b/a)\left[f(\infty)-f(0)\right]$$

Which pops up occasionally in analysis. For example, in the factorial sum just above we made use of the identity $\int_0^\infty \frac{e^{-t}-e^{-zt}}{t}dt = \ln(z)$ and in our first log-sine formula, we notice a suspicious $\frac{\cosh(0t)-\cosh(\alpha t)}{t}$ appearing in the integrand.

Your task is to give a proof of the identity by making use of the delightful trick $f(bt)-f(at) = t\int_a^b f'(ut)du$. Then, discuss how an arbitrary product $\prod_n \frac{b_n}{a_n}$, could, at least in theory, be expressed in the form of an integral, say something like $\ln\left(\prod_n \frac{b_n}{a_n}\right) = \int_0^\infty I(f,a,b,x,\dots)dx$.

Finally, Frullani makes somewhat frequent appearances in Integration Bees. Try to rearrange the following into an "insta-solve" application of the identity!

$$\text{[UC Berkeley 2021]} \quad \int_0^\infty \frac{\left(e^{3x}+1\right)-\left(e^x+1\right)}{\left(e^x+1\right)\left(e^{3x}+1\right)}\frac{dx}{x}$$

$$\text{[Silver Mock Exam]} \quad \int_0^\infty \frac{1}{\sqrt[3]{x^3+e}} - \frac{1}{\sqrt[3]{x^3+e^2}}dx$$

For many more bee examples, check out `https://www.youtube.com/watch?v=dwjR-kYDPkQ`

Finally, we conclude by translating two miscellaneous Mittag-Leffler-type expansions of trigonometric functions into the familiar Laplace-Transform form;

$$\pi\cot(\pi\alpha) = \frac{1}{\alpha} + \sum_{n\geq 1}\left(\frac{1}{\alpha+n}+\frac{1}{\alpha-n}\right) = \frac{1}{\alpha} + \int_0^\infty \frac{e^{-\alpha t}-e^{\alpha t}}{e^t-1}dt$$

$$\pi\tan(\pi\alpha) = \sum_{n\geq 0}\frac{2\alpha}{\left(n+\frac{1}{2}\right)^2-\alpha^2} = \sum_{n\geq 0}\int_0^\infty e^{-nt}e^{-\frac{1}{2}t}2\sinh(\alpha t)\,dt = \int_0^\infty \frac{e^{\alpha t}-e^{-\alpha t}}{e^t-1}\sqrt{e^t}dt$$

Since both of these series, as sums of rational functions, are perfectly phrased as a geometric series in the Laplace Transforms of $\mathcal{L}[e^{\pm nt}] = \frac{1}{s\mp n}$ and $\mathcal{L}[e^{-1/2t}\sinh(\alpha t)] = \frac{\alpha}{(s+1/2)^2-\alpha^2}$ respectively. There are plenty of more trigonometric identities like this, but that's the idea.

Exercise: How can you use $\frac{d}{d\alpha}\ln\sin(\pi\alpha)$ and $\frac{d}{d\alpha}\ln\sin(\pi\alpha+\frac{\pi}{2})$ to derive the above two identities from the opening $\ln\left(\frac{\sin(\pi\alpha)}{\pi\alpha}\right)$ example? In particular, derive the claimed Mittag-Leffler expansions by differentiating $\frac{d}{d\alpha}\sum_{n\geq 1}\ln\left(1-\frac{\alpha^2}{n^2}\right)$. Why is there only a $\frac{1}{e^t-1}$ "suffix" instead of the usual $\frac{1}{t}\frac{1}{e^t-1}$ term?

Post's Inversion

Given $F(s)$, the use of an inverse-Laplace transform table is almost surely the most familiar means of recovering $f(t)$. However, it may be surprising to know that several closed-form formulas for f in terms of F exist, and one that is particularly pertinent to the topic of "differentiating the Laplace Transform" is the following due to Emil Post:

$$f(t) = \lim_{k\to\infty}\frac{(-1)^k}{k!}\left(\frac{k}{t}\right)^{k+1}F^{(k)}\left(\frac{k}{t}\right)$$

Which indeed makes good use of derivatives of $F(s)$.

Exercise: Verify for yourself that the inversion formula really works for the case of $F(s) = \frac{1}{s-a}$! Start by using induction to show $F^{(k)}(s) = (-1)^k k! \frac{1}{(s-a)^{k+1}}$ and recall the limit $\lim_{k\to\infty} \left(1 - \frac{x}{k}\right)^{-k} = e^x$: it's quite the beautiful simplification.

To prove the above, it suffices to check only the $t = 1$ case, and this is due to the identity $\mathcal{L}[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right)$ which tells us that if $f(1) = \lim_{k\to\infty} \frac{(-1)^k}{k!} k^{k+1} F^{(k)}(k)$ then we can cleanly rescale this to reach any value for $f(a)$:

$$f(a) = \lim_{k\to\infty} \frac{(-1)^k}{k!} k^{k+1} \left[\frac{d^k}{ds^k} \frac{1}{a} F\left(\frac{s}{a}\right)\right]_{s\to k} = \frac{(-1)^k}{k!} k^{k+1} \frac{1}{a^{k+1}} F^{(k)}\left(\frac{k}{a}\right)$$

As desired. Now, by making use of the identity $(-1)^k F^{(k)}(s) = \int_0^\infty e^{-st} t^k f(t) dt$, we see that what Post's formula is really establishing is the following *filtration*;

$$f(1) = \lim_{k\to\infty} \frac{k^{k+1}}{k!} \int_0^\infty e^{-kt} t^k f(t) dt = \lim_{k\to\infty} \int_0^\infty \delta_k(t) f(t) dt$$

Where $\delta_k(t) = \frac{k^{k+1}}{k!} e^{-kt} t^k$ is a marvelous sequence of kernels converging to a *delta distribution* at $t = 1$: thereby extracting the quantity $f(1)$ from the integrand. It's a nice exercise to check that $\int_0^\infty \delta_k(t) dt = 1$, meaning that these do provide a sequence of *distributions*, but verifying that they converge to a Dirac spike at 1 is more challenging: see `https://www.rosehulman.edu/~bryan/invlap.pdf` for all the details.

I was interested in whether such a clean, limit-based technique could be used to invert other types of integral transforms, such as the Mellin Transform $\mathcal{M}[f(t)] = \int_0^\infty f(t) t^{s-1} dt$. However, I could not find such convenient densities δ_k as Post did that were well-defined on the infinite domain of the integral, that converged in the right way to a delta distribution, *and* which could be easily integrated in terms of clean derivatives of $\mathcal{M}$. I thought that $\delta_k(t) = \left(\frac{t\ln(t)}{t^2-1}\right)^k$ showed some promise in the case of Mellin, but the denominator is too hard to work with.

Still, one can easily "cheat" this problem entirely by throwing polynomials at the wall: it is a famous result that certain sequences of polynomials such as the *Laguerre Polynomials* $L_n(x) = \sum_{k=0}^n \binom{n}{k} \frac{(-1)^k}{k!} x^k$ multiply together to give the δ distribution, a property known as *completeness* of the family/sequence. Concretely, we'll be assuming the following identity as a fact;

$$\lim_{N\to\infty} \underbrace{\frac{1}{e} \sum_{n=0}^N L_n(1) L_n(t)}_{\delta_N(t)} = \delta(t-1)$$

This is a *formal* identity: in reality the above sum does *not* converge to anything sensible, only the sequence of integrals $\int \delta_N(t) f(t) dt$ converge when f is sufficiently quickly decaying as $t \to \infty$. In any case, by re-using our re-scaling trick, ie $f(t) \mapsto f(at)$, from above, we can argue that;

$$e^{-s} f(a) = \int_0^\infty f(at) e^{-st} \delta(t-1) dt = \lim_{N\to\infty} \frac{1}{e} \sum_{n=0}^N L_n(1) \sum_{k=0}^n \binom{n}{k} \frac{(-1)^k}{k!} \int_0^\infty f(at) t^k e^{-st} dt = \frac{1}{e} \sum_{n=0}^\infty L_n(1) \sum_{k=0}^n \binom{n}{k} \frac{a^{-k-1}}{k!} F^{(k)}\left(\frac{s}{a}\right)$$

Where this last integral is of course recognizable as a scaled derivative of the Laplace transform of $f(t)$. For example, should we attempt to invert $F(s) = \frac{1}{s-c}$ we can make use of one of Wikipedia's generating function characterizations of the Laguerre polynomials:

$$\sum_{n=0}^\infty L_n(t) z^n = \frac{1}{1-z} e^{-\frac{tz}{1-z}} \implies f(a) = e^s \frac{1}{e} \sum_{n=0}^\infty L_n(1) \sum_{k=0}^n \binom{n}{k} \frac{a^{-k-1}}{k!} \underbrace{\frac{(-1)^k k!}{\left(\frac{s}{a} - c\right)^{k+1}}}_{F^{(k)}(s/a)} = e^{s-1} \frac{1}{s-ac} \frac{1}{1-z} e^{-\frac{1z}{1-z}} = e^{ac}$$

Where $z = 1 - \frac{1}{s-ac}$ is the result of simplifying the innermost sum as the binomial $(1+z)^n$: in the end we correctly arrive at $f(t) = e^{ct}$ evaluated at the point $t \to a$!

> *Exercise:* Establish the identity $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{n!}{(n-k)!}\left(\frac{x}{n}\right)^k = \frac{1}{1-x}$ for $|x|<1$ using the Binomial Theorem and a Laplace Transform at $s=1/x$. This identity is at least mildly interesting since we've "reweighted" the geometric series $\lim_{n\to\infty}\sum_{k=0}^{n} x^k = \frac{1}{1-x}$ and got the same result.

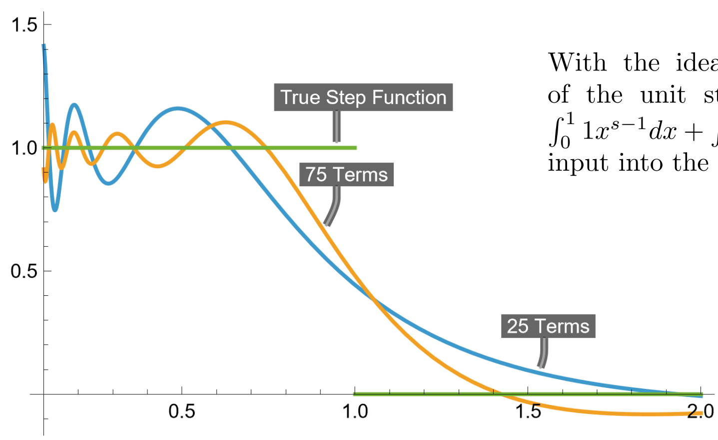
Clearly the above inversion formula is far, far more complex than Post's, but it does provide us with one curious idea: by crafting a bespoke δ-function from Laguerre polynomials L_n, we can undo an integral transform. This path leads to a wildly impractical formula for inverting the *Mellin* transform;

$$f(a) = \frac{c^{1-s}}{e^c}\sum_{n=0}^{\infty} L_n(c)\sum_{k=0}^{n}\binom{n}{k}\frac{(-1)^k}{k!}\left(\frac{c}{a}\right)^{(k+s)}\mathcal{M}(s+k)$$

Which just comes from expanding out a delta function, together with standard shift / sclae transformations, in terms of L_n:

$$f(a)c^{s-1} = \int_0^\infty f\left(\frac{a}{c}t\right)\delta(t-c)t^{s-1} = \frac{1}{e^c}\sum_{n\geq 0} L_n(c)\sum_{k=0}^{n}\binom{n}{k}\frac{(-1)^k}{k!}\int_0^\infty f\left(\frac{a}{c}t\right)t^{k+s-1}dt = \frac{1}{e^c}\sum_{n\geq 0} L_n(c)\sum_{k=0}^{n}\binom{n}{k}\frac{(-1)^k}{k!}(c/a)^{k+s}\mathcal{M}(k+s)$$

When I say the formula is *wildly* impractical this is no exaggeration: not only is $f(0)$ undefined but unlike with the Laplace transform, increasing the argument s generally causes $\mathcal{M}(s)$ to rapidly *diverge* rather than decrease, and so only after carefully setting $s\approx 1$, $c\approx 1/2$ and $\mathcal{M}(s)=1/s$ was I able to obtain anything even *remotely* recognizable:



With the idea being that the Mellin transform of the unit step function (plotted in green) is $\int_0^1 1x^{s-1}dx + \int_1^\infty 0x^{s-1}dx = \frac{1}{s}$, which matches our input into the inversion formula.

A Cosine Addendum

Although completely different in origin to Laplace Transforms, I have in my notes another intriguing integral-based identity for the binomial coefficient without logs and with cosine;

$$\mathcal{I}_{n,k} = \binom{n}{k} = 2^n\int_0^1 \cos(\pi x)^n\cos(\pi x(n-2k))\,dx$$

Which can be thought of as a delightful generalization of the more well-known *central* binomial integral identity (something which makes frequent appearaences in both analysis and integration bees):

$$\mathcal{I}_{2n,n} = 2^{2n}\int_0^1 \cos(\pi x)^{2n}\cos(\pi x(2n-2n))\,dx = 2^{2n}\int_0^1 \cos(\pi x)^{2n}\,dx = \binom{2n}{n}$$

Using Complex Analysis, specifically Cauchy's Formula, a proof of the above binomial identity requires no more than a few lines:

$$\binom{n}{k} = \mathrm{Coeff}(z^k)[(1+z)^n] = \frac{1}{2\pi i}\oint_{|z|=1}\frac{(1+z)^n}{z^{k+1}}dz = \int_0^1\frac{(1+e^{2\pi it})^n}{e^{2\pi it(k+1)}}e^{2\pi it}dt = 2^n\int_0^1 e^{\pi int}\left(\frac{e^{-\pi it}+e^{\pi it}}{2}\right)^n e^{-2\pi ikt}dt$$

With the idea being that we are extracting the coefficient of the kth power of z in the Laurent expansion of $(1+z)^n$, then taking the real part of the integral to turn exponentials into cosines. However, I think it is much more insightful to see how the $\cos(\pi x n - 2\pi k x)$ term *encodes* certain recursive / combinatorial information about $\binom{n}{k}$.

For example, by virtue of a sum-to-product formula, we can analytically construct Pascal's Identity $\binom{n-1}{k} + \binom{n-1}{k-1} = \binom{n}{k}$;

$$\mathcal{I}_{n-1,k-1} + \mathcal{I}_{n-1,k} = 2^{n-1} \int_0^1 \cos(\pi x)^{n-1} \left[\cos(\pi x(n-1-2k)) + \cos(\pi x(n+1-2k))\right] dx$$

$$= 2^{n-1} \int_0^1 \cos(\pi x)^{n-1}\, 2\cos(\pi x(n-2k)) \cos(\pi x(1))\, dx = \mathcal{I}_{n,k}$$

Similarly, using integration by parts we can form an additional $\binom{n-1}{k-1}\frac{n}{k} = \binom{n}{k}$ identity;

$$\frac{n}{k}\mathcal{I}_{n-1,k-1} = \frac{n}{k} 2^{(n-1)} \int_0^1 \cos(\pi x)^{(n-1)} \cos(\pi x(n-2k+1))\, dx$$

$$= \frac{n}{k} 2^{(n-1)} \int_0^1 \cos(\pi x)^{n} \cos(\pi x(n-2k))\, dx - \frac{n}{k} 2^{(n-1)} \int_0^1 \cos(\pi x)^{(n-1)} \sin(\pi x(n-2k)) \sin(\pi x)\, dx$$

$$= \frac{n}{k} 2^{(n-1)} \int_0^1 \cos(\pi x)^{n} \cos(\pi x(n-2k))\, dx - \frac{(n-2k)}{k} 2^{(n-1)} \int_0^1 \cos(\pi x)^{n} \cos(\pi x(n-2k))\, dx$$

$$= \mathcal{I}_{n,k}$$

And together, these two properties of the binomial coefficient provide a two-variable induction for $\forall n, k \in \mathbb{N}$, drawing off of a base case like $\binom{0}{k} = 2^0 \int_0^1 \cos(\pi x)^0 \cos(2\pi k x) dx = \begin{cases} 1 & k = 0 \\ 0 & k > 0 \end{cases}$ to seed the edges of Pascal's triangle.

Exercise: For $x \in [-1/2, 1/2]$ and for n sufficiently large, $\cos(\pi x)^{2n}$ can be nicely approximated by a Gaussian $e^{-n\pi^2 x^2}$: convince yourself at least intuitively that both curves have the same general shape and behavior as $n \to \infty$. Your task is to get an asymptotic estimate for the size of $\binom{2n}{n}$ using this and our integral formula above:

$$\frac{1}{4^n}\binom{2n}{n} = \int_0^1 \cos(\pi x)^{2n}\, dx = 2\int_0^{\frac{1}{2}} \cos(\pi x)^{2n}\, dx \approx 2\int_0^{\frac{1}{2}} e^{-n\pi^2 x^2} dx \approx 2\int_0^{\infty} e^{-n\pi^2 x^2} dx$$

Who could have expected that the growth of $\binom{2n}{n}$ would involve a π in it?